Network Synthesis: Foster and Cauer form Realization	
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Aim:

To Verify the Network Synthesis: Foster and Cauer form Realization

Theoretical Calculation:

Circuit Diagram:

Question 1. Determine the foster I and foster II for the given Impedence.

Qu 1

Foster-I

$$Z(s) = \frac{4(s^2+1)(s^2+16)}{s(s^2+4)} = \frac{A}{s} + \frac{Bs+C}{(s^2+4)} + Ds$$

$$s=0 \longrightarrow \boxed{A=16}$$

$$= 4s^4 + 64s^2 + 4s^2 + 64 = 16s^2 + 64 + Bs^2 + Cs + (Ds^2)(s^2+4)$$

$$= 16s^2 + 64 + Bs^2 + Cs + Ds^4 + 4Ds^2$$

$$4s^4 = Ds^4$$

$$\boxed{D=4}$$

$$\cancel{16s^2 + 4s^2 = 64s^2 + 4s^2}$$

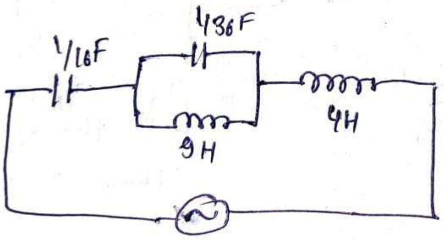
$$\Rightarrow (64+4)s^2 = (16+B+4D)s^2$$

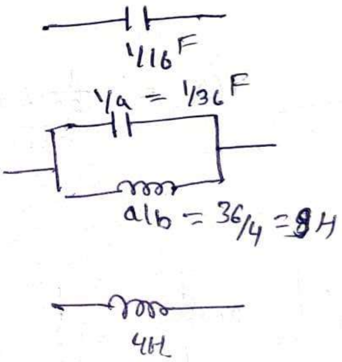
$$68-16 = B+4\times4 \Leftarrow \Rightarrow 68-16 = B+4D \Rightarrow \boxed{\cancel{B=52}}$$

$$52-16 = B \qquad Cs=0 = \boxed{C=0}$$

$$\boxed{B=36}$$

$$Z(s) = \frac{16}{s} + \frac{36s+0}{s^2+4} + 4s$$





Foster – II

$$Z(s) = \frac{4(s^2+1)(s^2+16)}{s(s^2+4)} \Rightarrow Y(s) = \frac{s(s^2+4)}{4(s^2+1)(s^2+16)}$$

$$Y(s) = \frac{s(s^2+4)}{4(s^2+1)(s^2+16)} = \frac{As+B}{s^2+1} + \frac{Cs+D}{s^2+16}$$

$$\Rightarrow s(s^2+4) = 4\left[(As+B)(s^2+16) + (Cs+D)(s^2+1)\right]$$

~~$s=0$ / = / 0 ≠ $A[$ / $B \times 16$ / +~~

$$s^3 + 4s = 4\left[\underline{As^3} + \underline{16As} + Bs^2 + 16B + \underline{Cs^3} + \underline{Cs} + Ds^2 + D\right]$$

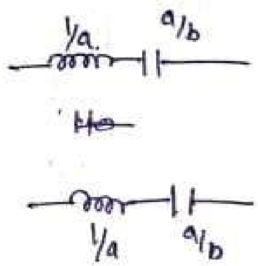
$$s^3 = As^3 + Cs^3 \qquad \frac{4s}{4} = 16As + Cs$$

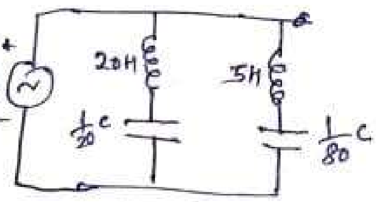
$$\boxed{\tfrac{1}{4} = A + C} \; -(1) \qquad \boxed{1 = 16A + C} \; -(2)$$

$$\boxed{\cancel{A = \tfrac{1}{5}}} \quad \cancel{C = \tfrac{4}{5}}$$

$$\boxed{A = \frac{1}{20}} \quad \boxed{C = \frac{1}{5}} \quad \boxed{B = 0} \quad \boxed{D = 0}$$

$$Y(s) = \frac{\frac{1}{20}s}{s^2+1} + \frac{\frac{1}{5}s}{s^2+16}$$





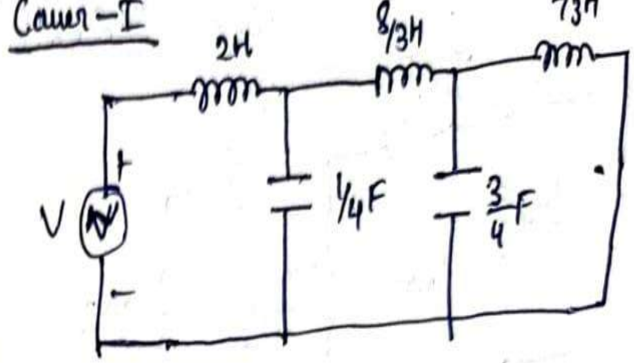
$$\frac{a}{b} = \frac{\frac{1}{20}}{\frac{1}{1}} = \frac{1}{20}$$

$$\frac{a}{b} = \frac{\frac{1}{5}}{\frac{16}{1}} = \frac{1}{16 \times 5}$$

Question 2 calculate the Cauer I and Cauer II for given Impedance.

Que 2

Cauer – I



$$Z(s) = \frac{2s^5 + 12s^3 + 16s}{s^4 + 4s^2 + 3}$$

$s^4 + 4s^2 + 3 \,\big)\, 2s^5 + 12s^3 + 16s \,\big(\, 2s \leftarrow Z(s), \; L = 2H$

$2s^5 + 8s^3 + 6s$

$4s^3 + 10s \,\big)\, s^4 + 4s^2 + 3 \,\big(\, \frac{1}{4}s \leftarrow Y(s), \; C = \frac{1}{4}F$

$s^4 + \frac{10}{4}s^2$

$\frac{3}{2}s^2 + 3 \,\big)\, 4s^3 + 10s \,\big(\, s\frac{2}{3} \times 4 \rightarrow \frac{8}{3} Z(s), \; L = \frac{8}{3}H$

$4s^3 + 8s$

$2s \,\big)\, \frac{3}{2}s^2 + 3 \,\big(\, \frac{1}{2}\frac{3}{2}s \rightarrow Y(s) \Rightarrow \frac{3}{4}F = C$

$\frac{3}{2}s^2 + \cancel{0}$

$3 \,\big)\, 2s \,\big(\, 2 \times \frac{1}{3}s \rightarrow Z(s)$

$2s$ $\qquad L = \frac{2}{3}H$

$\times\times$

Cauer-2

Qu. 2

$$Z(s) = \frac{2s^5+12s^3+16s}{(s^4+4s^2+3)} \Rightarrow Y(s)\ \frac{s^4+4s^2+3}{2s^5+12s^3+16s}$$

$$Y(s) = \frac{3+4s^2+s^4}{16s+12s^3+2s^5} ✓$$

$16s+12s^3+2s^5 \,)\ 3+4s^2+s^4 \ (\ \frac{3}{16s} \rightarrow Y(s) = L = \frac{16}{3}H$

$3+\frac{9}{4}s^2+\frac{3}{8}s^4$

$\frac{3}{16}\times 12 = \frac{9}{4}$

$\frac{7}{4}s^2+\frac{5}{8}s^4 \,)\ 16s+12s^3+2s^5 \ (\ \frac{4\times16}{7s} \rightarrow Z(s)\quad C=\frac{7}{16\times4}$

$16s+\frac{40}{7}s^3$

$\frac{2\times3}{16}$

$\frac{4\times16}{7}\times 12$

$\frac{44}{7}s^3+2s^5 \,)\ \frac{7}{4}s^2+\frac{5}{8}s^4 \ (\ \frac{7}{44s}\times\frac{7}{4} \rightarrow Y(s)$

$\frac{7}{4}s^2+\frac{49}{88}s^4 \qquad L=\frac{44\times4}{7\times7}H$

$\frac{3}{44}s^4 \,)\ \frac{44}{7}s^3+2s^5 \ (\ \frac{44}{3s}\times\frac{44}{7}$

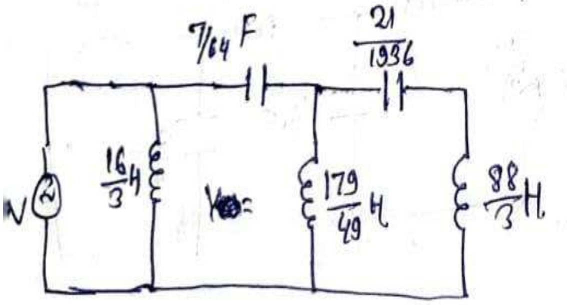
$Z(s)$

$C=\frac{3\times7}{44\times44}F \qquad \frac{44}{7}s^3$

$2s^5 \,)\ \frac{3}{44}s^4 \ (\ \frac{1}{2s}\frac{3}{44}$

$Y(s)$

$L=\frac{2\times44}{3}H$



Question 3. Determine the foster I for the given Impedance.

Que 3

$$Z(s) = \frac{5(s+1)(s+4)}{(s+3)(s+5)} = \frac{A}{(s+3)} + \frac{B}{s+5} + C$$

Foster-I

$$A = \frac{5\times(-2)\times(1)}{(2)} = \frac{-10}{2} = -5 \quad \text{(Negative)}$$

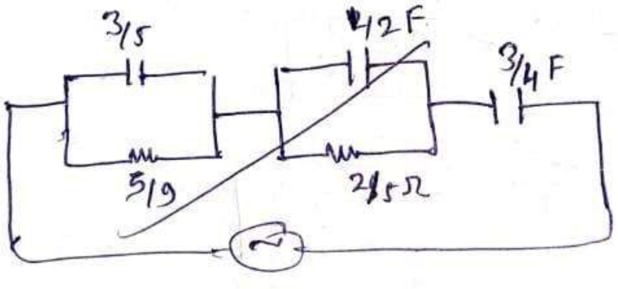
$$\frac{Z(s)}{s} = \frac{5(s+1)(s+4)}{s(s+3)(s+5)} = \frac{A}{(s+3)} + \frac{B}{(s+5)} + \frac{C}{s}$$

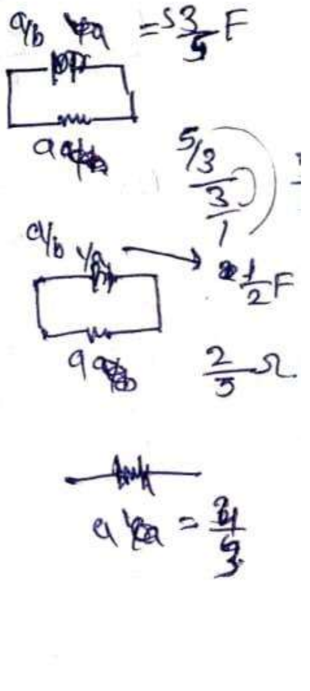
$$A = \frac{5\times(-2)\times(1)}{2\times(-3)} = \frac{10}{6} = \frac{5}{3}$$

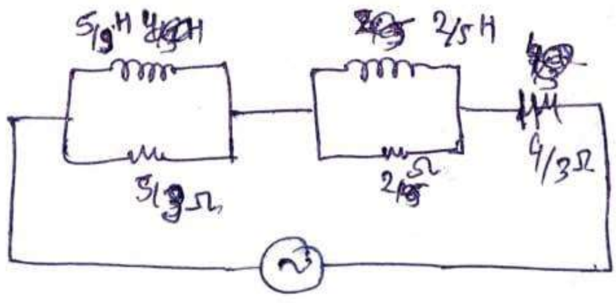
$$B = \frac{5(-4)(-1)}{(-2)(-5)} = \frac{20}{10} = 2$$

$$C = \frac{5\times1\times4}{3\times5} = \frac{20}{15} = \frac{4}{3}$$

$$Z(s) = \frac{s\,5/3}{s+3} + \frac{s\,2}{(s+5)} + \frac{\cancel{s}\,4/3}{\cancel{s}}$$







Question 4 calculate the Cauer I for given Impedance.

Que 4 $\quad Z(s) = \dfrac{(s+8)(s+4)}{(s+2)(s+6)} = \dfrac{s^2+4s+8s+32}{s^2+2s+6s+12} = \dfrac{s^2+12s+32}{s^2+8s+12}$

Cauer-I

$Z(s) = \dfrac{s^2+12s+32}{s^2+8s+12}$ $\quad \cancel{Y(s) = s^2+8s}$

$s^2+12s+32 \,\big)\, s^2+8s+12 \,\big(\, 1$

$s^2+12s+32$

$\overline{-4s+}$

$s^2+8s+12 \,\big)\, s^2+12s+32 \,\big(\, 1$

$s^2+8s+12$

$4s+20 \,\big)\, s^2+8s+12 \,\big(\, 4s$

$4s^2+20s$

$s^2+8s+12 \,\big)\, s^2+12s+32 \,\big(\, 1 \rightarrow Z(s)$ $\quad 1\Omega$

$s^2+8s+12$

$4s+20 \,\big)\, s^2+8s+12 \,\big(\, \frac{s}{4}$ $\quad Y(s) \quad C = \frac{1}{4}F$

s^2+5s

$3s+12 \,\big)\, 4s+20 \,\big(\, \frac{4}{3}$ $\quad Z(s) \quad 4/3\,\Omega$

$4s+16$

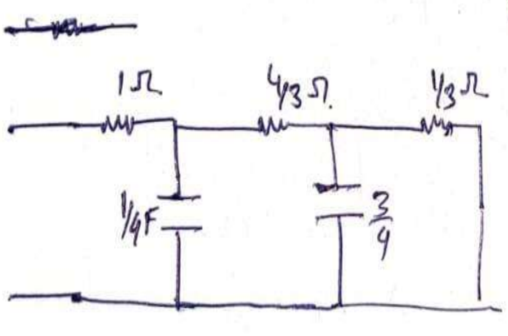
$4 \,\big)\, 3s+12 \,\big(\, \frac{3s}{4}$ $\quad Y(s) \quad C = \frac{3}{4}F$

$3s$

$12 \,\big)\, 4 \,\big(\, \frac{4}{12}$ $\quad Z(s) \quad 1/3\,\Omega$

4

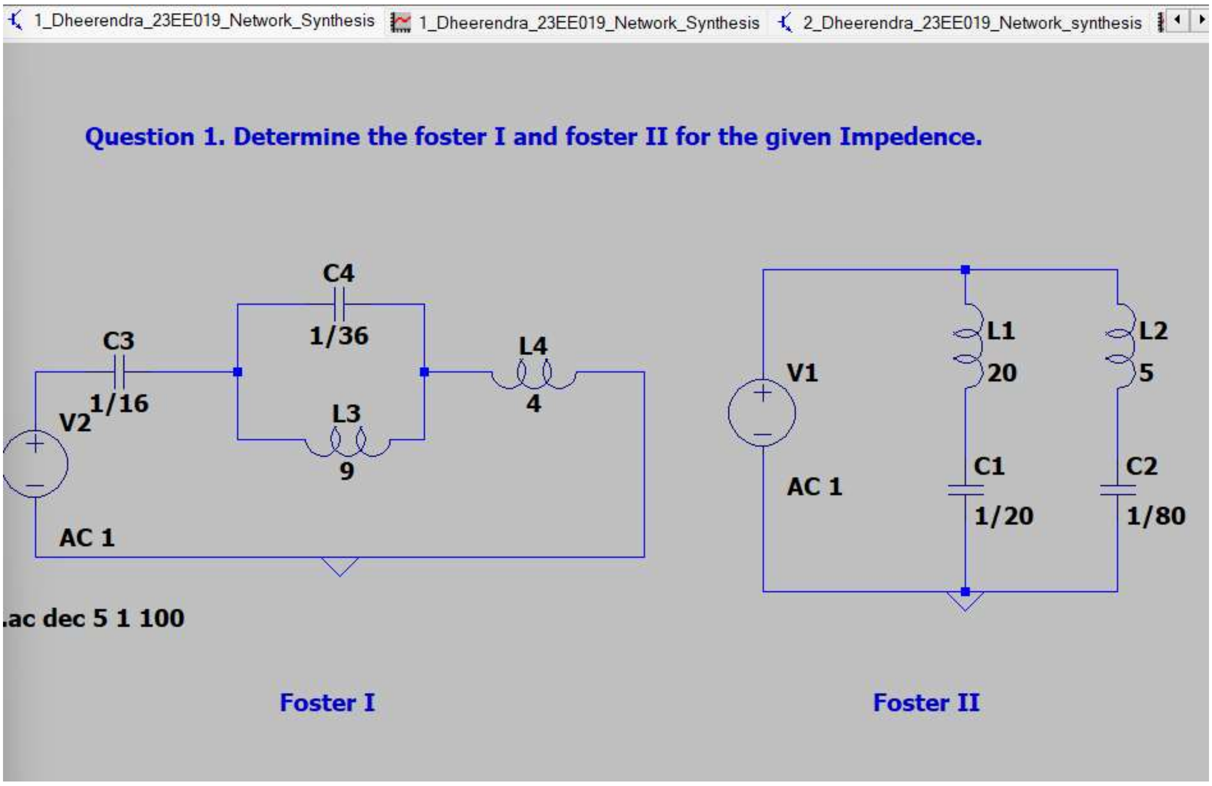
$\overline{\times}$



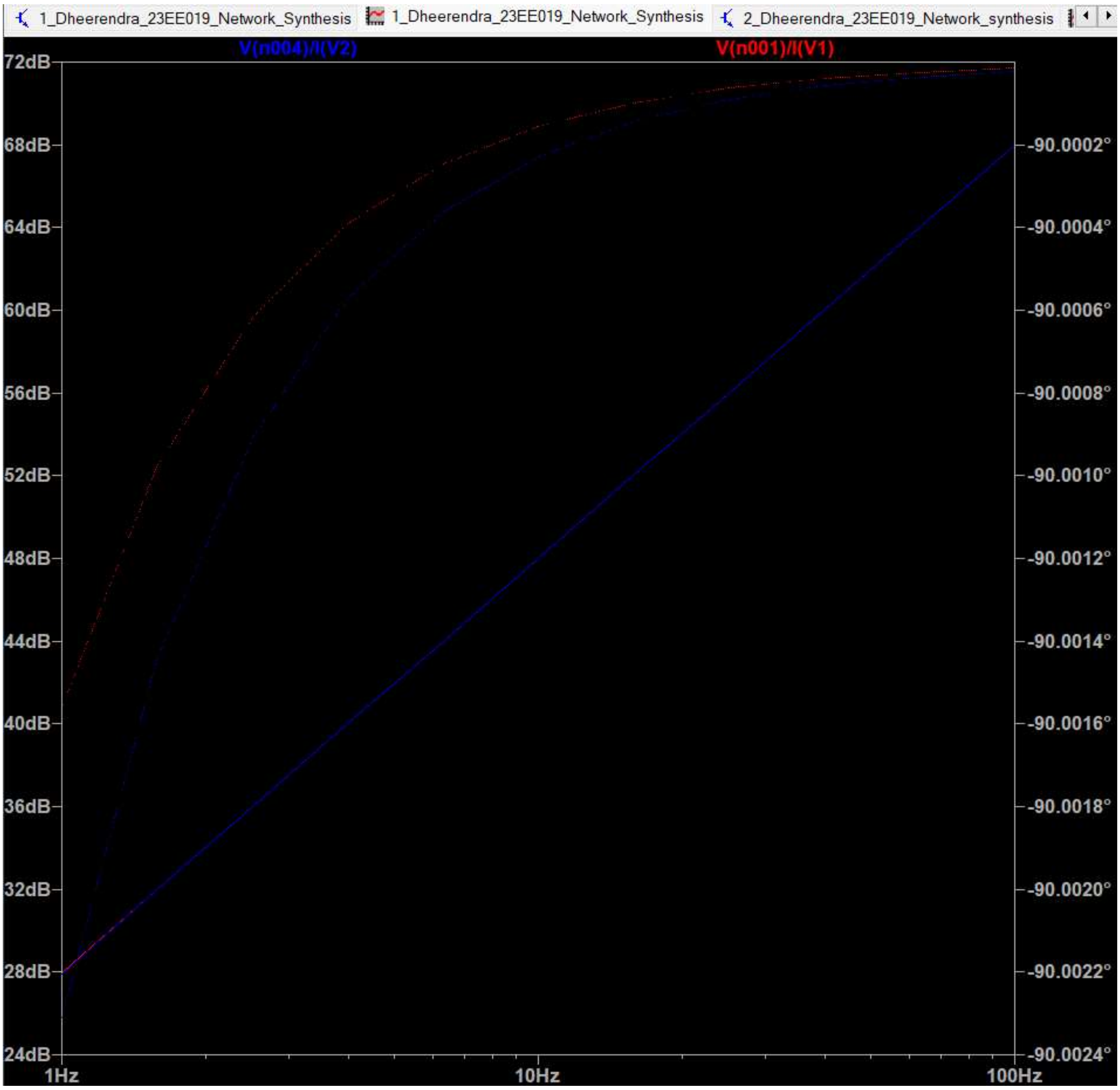
Verification by Simulation:

Question 1. Determine the foster I and foster II for the given Impedence.

Circuit Diagram:

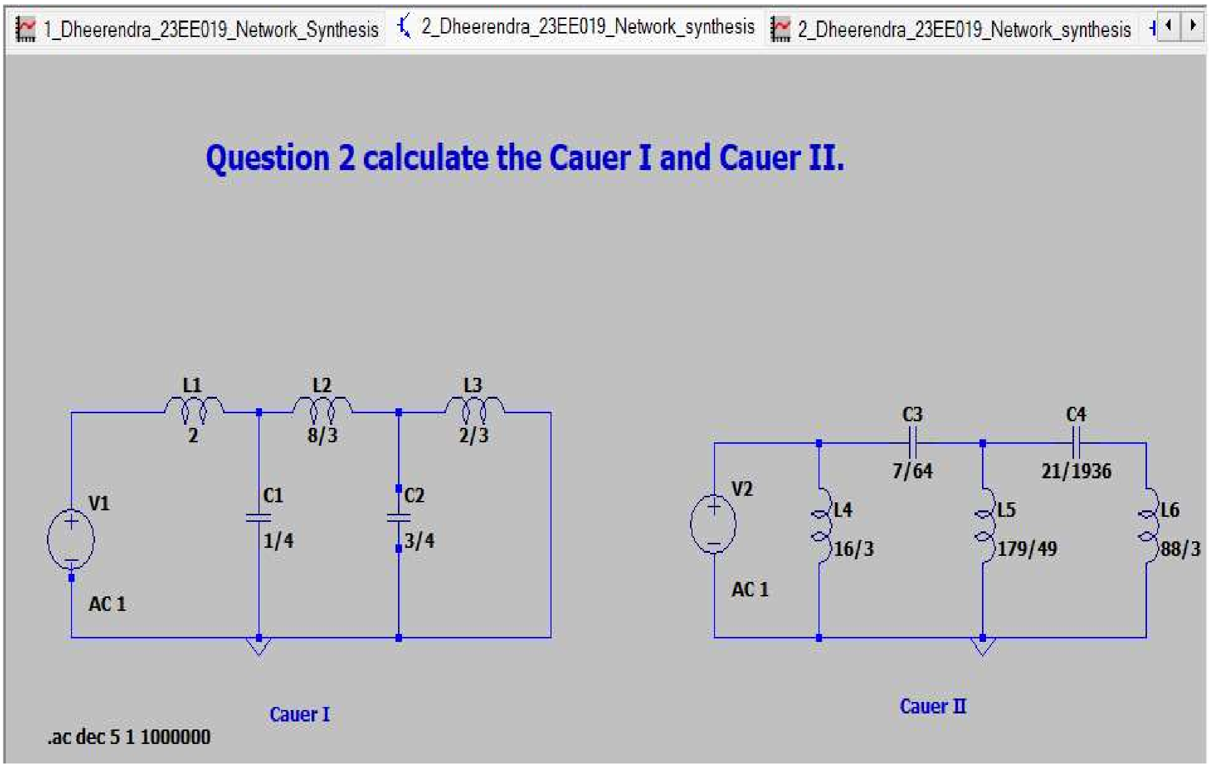


Plot of Voltage/Current:



Question 2 calculate the Cauer I and Cauer II.

Circuit Diagram:



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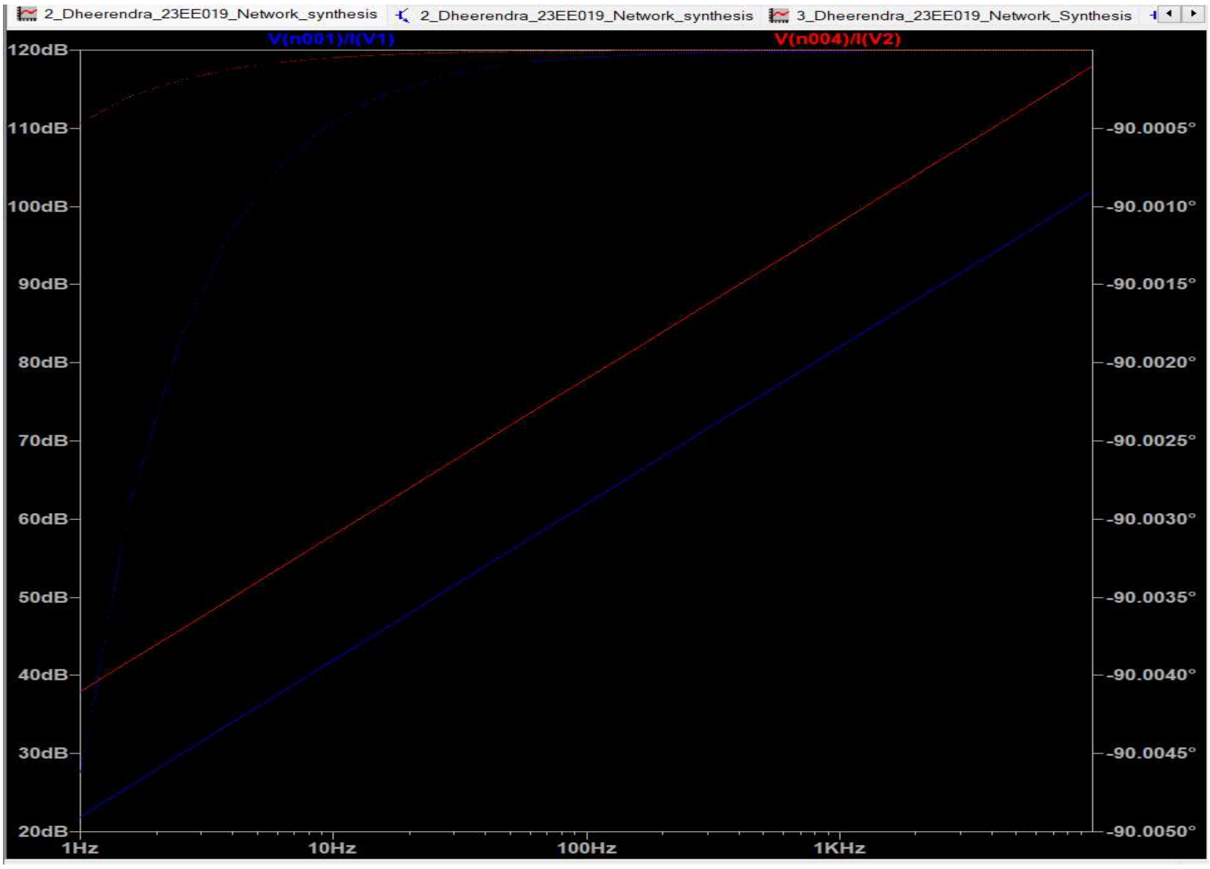
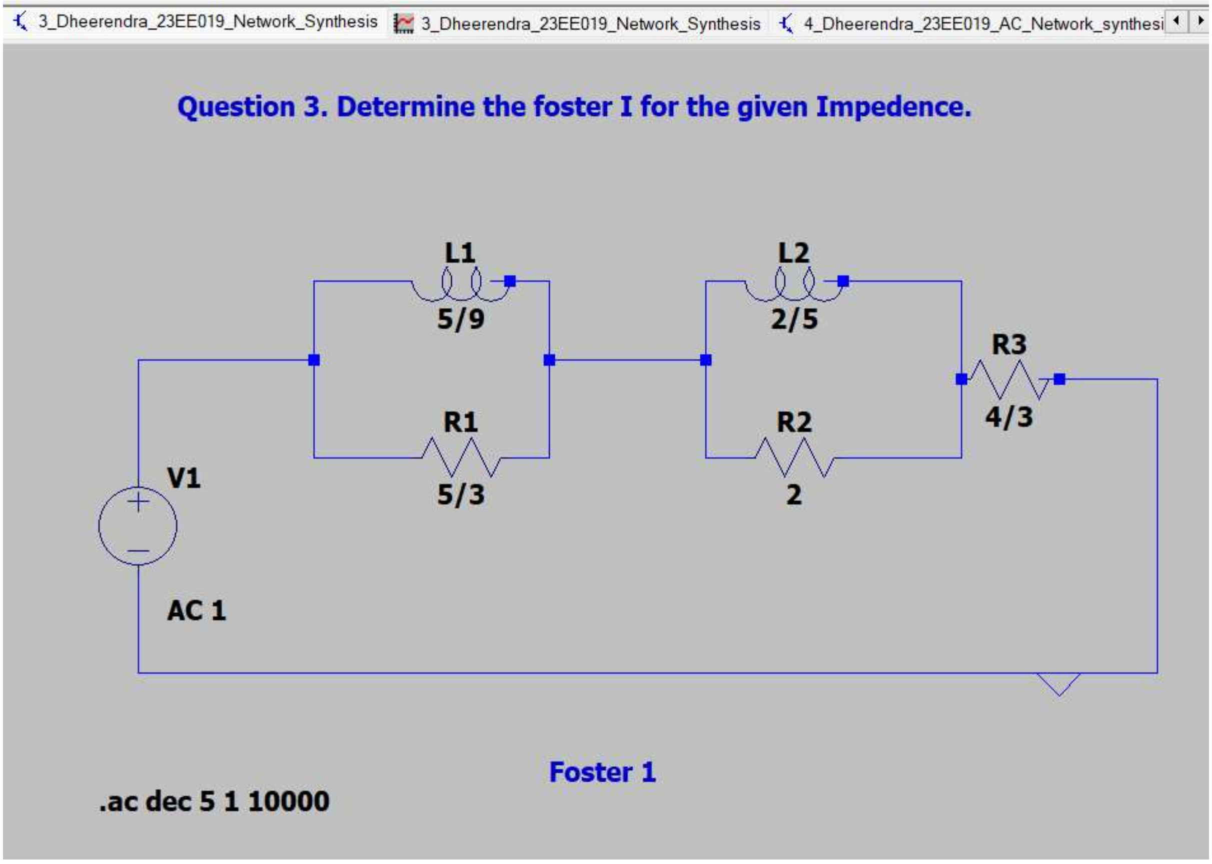
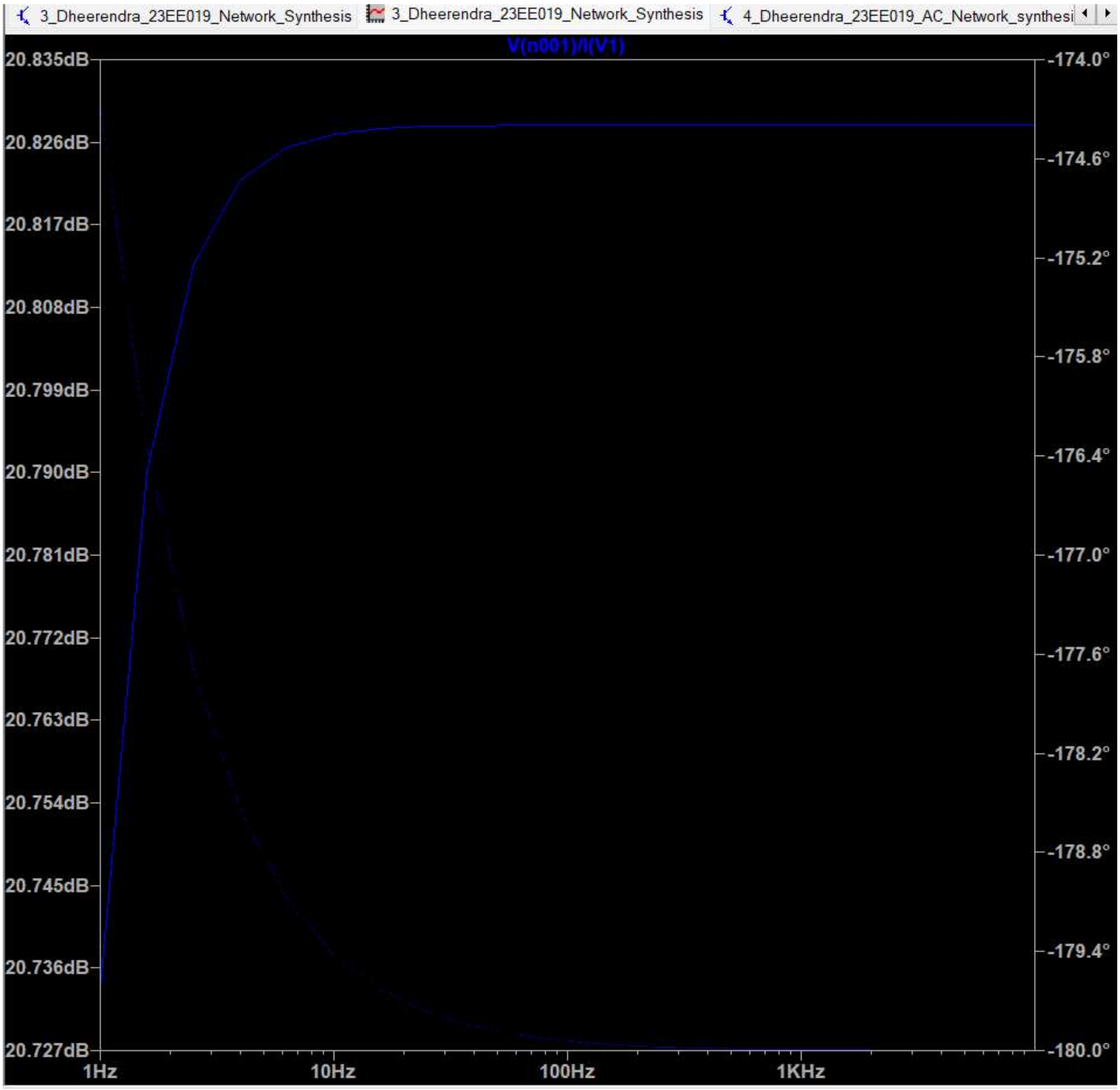


Question 3. Determine the foster I for the given Impedance.

Circuit Diagram:

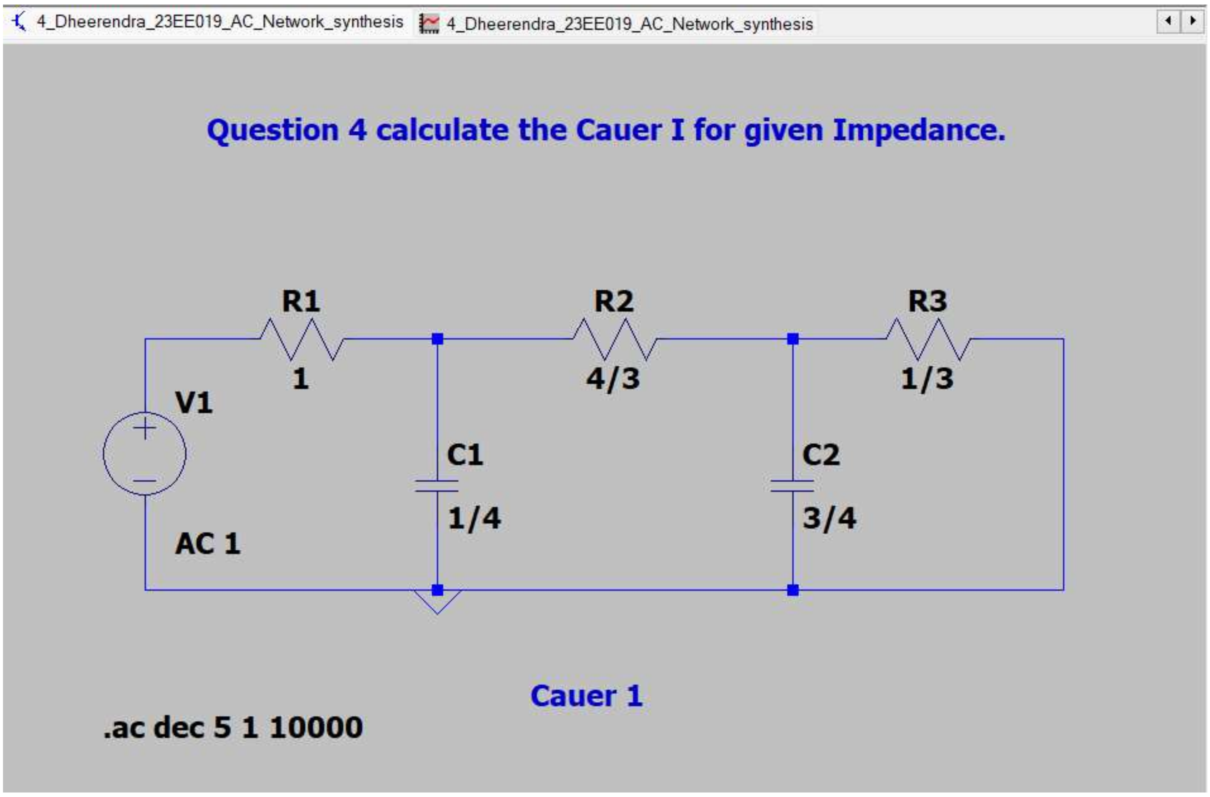


Plot of Voltage/Current:

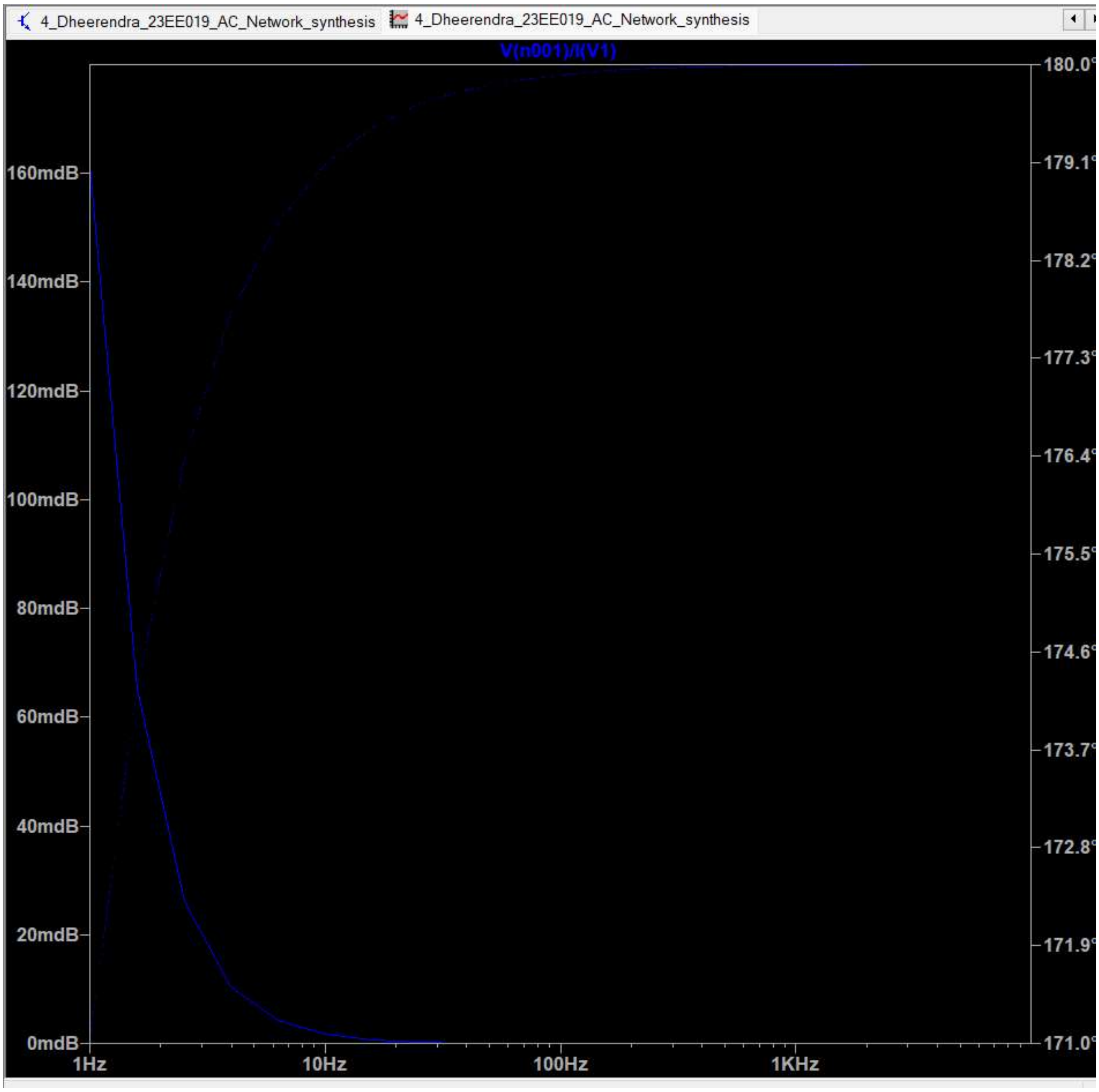


Question 4 calculate the Cauer I for given Impedance.

Circuit Diagram:



Plot of Voltage/Current:



Results and Conclusion:

As we see that the theoretical values are approximately equal to the practical values hence Network synthesis Foster I and Foster II and Cauer I and Cauer II are verified.